# Krishna Teja

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## EDUCATION

| Institution | Degree / Board | Passing Year | Score / CGPA |
|---|---|---|---|
| Indian Institute of Technology Hyderabad | B.Tech | 2027 | 8.28 |
| Suguna PIP School, Kalapatty Road, Coimbatore | Class XII (CBSE) | 2019 | 93.00% |
| Suguna PIP School, Kalapatty Road, Coimbatore | Class X (CBSE) | 2017 | 9.60% |

## EXPERIENCE

### SDE Internship at Amazon

- **Enhanced Tweety**, Amazon's internal canary management tool for database engine releases, **improving deployment reliability across AWS regions**.
- **Reduced Lambda cold-start latency by 90%** by enabling and testing Lambda SnapStart
- **Reduced list canaries API payload to below 1 MB** by implementing **server-side pagination** and optimizing client-side rendering for paginated responses
- **Debugged and resolved deployment blockers,** enabling Tweety rollout to 2 new AWS regions.

### Virtual Internship at Samsung Prism

- Built a Python-based ML model to **forecast network alarms** for the Network Fault Management System.
- **Created an alarm dataset** partially conforming to the TM Forum TMF642 alarm specification.
- **Trained and tested** Logistic Regression, K-Means, and MLP models in scikit-learn, **selected the best-performing model per alarm attribute,** and stored results in MongoDB Atlas.

## PROJECTS

### Optimizing Knowledge Graph-based Data Store for Network Digital Twins

- Built a Java Spring Boot application that maintains a **live Knowledge Graph of network topology and forwarding rules** in Neo4j, **processing SDN controller events** through an event-driven multiprocessing architecture
- **Implemented content-addressable node reuse** (Merkle DAG) to eliminate duplicate nodes, achieving **5-85x reduction in graph size** and **85-97% faster event processing** vs baseline
- **Stress-tested on K-Fat tree topologies (10, 12)**, handling 600-1000 events/second with varying packet routing conditions
- Github: Krishna-Teja732/EventDrivenKGBuilder

### Mock Transactions Website

- Built a **containerized PHP/MySQL web application** demonstrating production-grade security: **CSRF token validation**, **XSS prevention** via input sanitization and output encoding, and **RCE mitigation** via MIME validation and safe image processing.
- Configured Apache SSL, PHP-FPM session security, and implemented authentication; deployed via Docker Compose for reproducible testing.
- Github: Krishna-Teja732/NetworkSecurityProject

### Image Classification Model Selector

- Built an **automated TensorFlow Hub model selection tool** that benchmarks pre-trained models on custom datasets and ranks them by accuracy, **streamlining transfer learning for multiclass image classification.**
- N. Arulanand, D. Kamalraj, and **Krishna Teja**, "Image Classification Model Selector", *J. Innov. Image Process.*, vol. 4, no. 4, pp. 299-315, Jan. 2023, doi: 10.36548/jiip.2022.4.007
- Github: Krishna-Teja732/Img_classification_model_selector

### Blogging Website using NodeJS

- Built a full-stack blogging platform with Node.js, PostgreSQL, and Handlebars, **implementing personalized content recommendations via cosine similarity** on user interest tags and blog tags.
- GitHub: kamalraj-ten/blog_website

## PUBLICATIONS

**Integrating Network State Verification with Knowledge Graph-based Digital Twin for SDN, D. Shinde, S. Sravan V, Krishna Teja, D. Raj R, K. Kataoka | IEEE NFV-SDN 2025**

- Co-authored research defining network state verification using Knowledge Graph-based digital twins. Reviewed data-plane and SDN control-plane verifiers, contributing to technical clarity.
- DOI: 10.1109/NFV-SDN66355.2025.11349405 | Pages 1-7

## TECHNICAL SKILLS

Languages: Java, Python, *C++, PHP, Neo4j Cypher, SQL, *Lua

Frameworks & Tools: Spring Boot, Docker, Git, AWS Lambda, Apache HTTPD, Node.js

Databases & Storage: Neo4j, MySQL, MongoDB

## RELEVANT COURSES

Advanced Computer Networks, Advanced Data Structures and Algorithms, Foundations of Machine Learning, Network Security, Network Engineering, Fraud Analytics, Linear Optimization, Deep Learning

## HOBBIES

I play video games such as Black Myth: Wukong, Sekiro, Elden Ring, Stray, and Little Nightmares.

Outside of gaming, I like jogging and tinkering with my Arch Linux setup on my laptop.